STRICT RECIPROCITY ON LG AND OG, AND THE HALF-INTEGER ODD-ZERO LAW (PROVE 254)

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Proof file	`grandpa-rick/rick-research` commit `a2dff63`, `proofs/2026-10-25-crownF-strict-reciprocity.md` `https://github.com/grandpa-rick/rick-research/blob/a2dff638efdd1ff2480aa131b80449cdacfe959c/proofs/2026-10-25-crownF-strict-reciprocity.md`
Registry	`proofs/registry/q488-converse.json`: nodes `q488-oc-Q30-KLG-strict-fillings`, `q488-oc-HZ-half-integer-law` (trust: proved, given (G0), (G1), BR first-hand).
Referee	§3.3 and §9.3 are hand-written case analyses, re-read once. Please referee those two.

Grades come from my registry ladder hunch < sketched < computed < checked-sober < proved. Every claim below carries one. The file name of the source markdown carries the date PROVE.md asked for (2026-10-25); the work was done 2026-10-24.

0. Verdict

item	statement	grade
Lemma R (reduction)	#strict order-preserving maps $P \to \{e < o\}$ with $D \mapsto e$ is 0 if P has a 3-chain or a non-minimal element of D, else $2^{\#\text{isolated elements} \notin D}$	proved (§2)
Theorem Q30	LG, dimension-indexed: $[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1 = \sum_\rho c_\rho [\mathcal{O}_{X_\rho}]$ with $c_\rho = (-1)^{\lvert\theta\rvert - 1} V(\theta)$, $V(\theta) = \#$strict order-preserving $\theta \to \{e < o\}$ with diagonal $\mapsto e$, for every $\rho \subsetneq \kappa$ (and $c_\rho = 0$ for $\rho \not\subseteq \kappa$)	**proved given BR Cor 5.4 + Thm 5.5** (both proved in BR, read first-hand), §3
Theorem Q30_{OG}	OG: $c_\rho = (-1)^{\lvert\theta\rvert - 1}$ if θ is a nonempty antichain (= nonempty set of maximal boxes of P_κ), else 0	proved given BR Lemma vanishB (I file Lemma 2), §3.4
Theorem SR_{LG}	$(-1)^{\lvert\kappa\rvert} H_\kappa(-j) = \#\{$strict order-preserving $P_\kappa \to [2j-1]$, diagonal values even$\}$, all $j \geq 1$	**proved given (G0), (G1), (BR)**, §4
Theorem SR_{OG}	$(-1)^{\lvert\kappa\rvert} H_\kappa^{\text{OG}}(-j) = \bar{\Omega}(P_\kappa, j-1) = \#\{$strict order-preserving $P_\kappa \to [j-1]\}$, all $j \geq 1$	**proved given (G0), (G1), (BR)**, §5
Corollary (Theorem I, combinatorial proof)	zero window, sign and monotonicity of Theorem I read off from the count	proved, §6

Date: 2026-10-24.

Theorem HZ (odd-zero law, both halves)	$H_\kappa(-a-\frac{1}{2})=0 \iff a \le \lceil r(\kappa')/2\rceil$; sign $(-1)^{\lvert\kappa\rvert-\kappa_1}$ for $2a\le\kappa_1$, $(-1)^{\lvert\kappa\rvert}$ for $2a\ge\kappa_1$; $\lvert\text{value}\rvert \ge 2^{-\kappa_1}N_{\kappa'}(a)$	**proved given (G0), (G1), (BR) + Thm A**, §9

The point: $(\star)$ of the I file is a positive recursion with a trivial anchor. Unrolled to the anchor, it *is* the counting formula. The only nontrivial input is that the one-step coefficient $|c_\rho|$ counts one "layer" $\{e<o\}$, and that is Q30. $\mathrm{SR}_{\mathrm{LG}}$ no longer depends on the LS-parity model. $\mathrm{SR}_{\mathrm{OG}}$ no longer depends on SMT/Stanley reciprocity.

1. Setup

As in the I file (`proofs/2026-10-24-integer-window-and-OG-twin.md` §1; "the Z file" is `proofs/2026-10-24-Z-by-hyperplane-induction.md`. $X=\mathrm{LG}(m,2m)$ or $\mathrm{OG}(m,2m+1)$, L the ample generator, $X_\kappa := X^{\kappa^\vee}$ of dimension $|\kappa|$, $H_\kappa(j)=\chi(X_\kappa,L^j)$. $P_\kappa = D(\kappa)$ is the shifted diagram: boxes (a,b) with $0\le a\le b$, 0-indexed, diagonal $=\{a=b\}$. Order: $x\le y$ iff x is weakly NW of y (product order). For $\rho\subseteq\kappa$, $\theta=D(\kappa)\setminus D(\rho)$ is an up-set (order filter) of P_κ, and every up-set arises this way for a unique strict ρ (I file §2, Fact).

Inputs: (G0) $\mathcal{O}^1 = 1-[L^{-1}]$; (G1) Demazure character formula; (BR) Buch–Ravikumar arXiv:1005.2605, read first-hand from the v1 LaTeX source. Used here: §5 eq. (11) + **Cor 5.4** ($c^\nu_{\lambda,p}=\mathcal{C}(\nu/\lambda,p)$ for $\lambda\subset\nu$) and **Thm 5.5** (the arm recursion for $\mathcal{C}$), both proved in BR. We do **not** use Cor 5.6 (KLG tableaux), whose proof BR leave to the reader; §3.3 below is that proof at $p=1$. §4 OG: Lemma vanishB (proved in BR).

The I file's $(\star)$: with $s_j(\kappa):=(-1)^{|\kappa|}H_\kappa(-j)$,

$$(\star)\qquad s_{j+1}(\kappa)=s_j(\kappa)+\sum_{\rho\subsetneq\kappa}|c_\rho|\,s_j(\rho)\qquad\text{for every integer } j,$$

valid on LG and OG, given (G0) and the sign rule $c_\rho=(-1)^{|\theta|-1}|c_\rho|$ (re-proved below on LG as part of Q30). Anchor: $s_1(\kappa)=0$ for $\kappa\neq\emptyset$ (I file Lemma 1, given (G1)); $s_j(\emptyset)=1$ ($X_\emptyset$ is a point).

2. Lemma R (counting $\{e<o\}$ maps)

Let P be a finite poset and $D\subseteq P$. Let $V(P,D)=\#\{f:P\to\{e<o\} : x<y\Rightarrow f(x)<f(y),\ f(D)\subseteq\{e\}\}$.

Lemma R. *[proved] $V(P,D)=0$ if P has a chain $x<y<z$, or if some $d\in D$ is not minimal. Otherwise $V(P,D)=2^i$, where i is the number of isolated elements (comparable to nothing else) not in D.*

Proof. A 3-chain needs three distinct values. If $x<d\in D$ then $f(x)<f(d)=e$ is impossible. Otherwise P has height ≤ 2. A non-isolated x is then either non-maximal ($x<y$ for some y, forcing $f(x)=e$, $f(y)=o$) or non-minimal (forcing $f(x)=o$), not both (that would be a 3-chain). These forced values satisfy every relation $x<y$ (x non-maximal $\mapsto e$, y non-minimal $\mapsto o$). A non-isolated $d\in D$ is minimal, hence non-maximal, hence forced to e, consistent with D. An isolated element is free (2 choices) unless it is in D (1 choice). □

In P_κ: a diagonal box (c,c) of an up-set θ is non-minimal in θ iff $(c-1,c)\in\theta$. (The boxes below (c,c) in P_κ are those weakly NW of it, i.e. (a,b) with $a\le b\le c$, $a\le c$. If one of them, $(a,b)\neq(c,c)$, lies in θ, then $(c-1,c)$, which lies between (a,b) and (c,c) when $a\le c-1$, $b\le c$, lies in θ since θ is an up-set. And $a=c$ forces $b=c$.)

3. Theorem Q30 (LG K-Pieri at $p=1$)

3.1 Translation to BR's side. $c_\rho = c^{\rho^\vee}_{\kappa^\vee,1}$ (dimension vs codimension indexing). If $\rho\not\subseteq\kappa$ then $c_\rho=0$ (I file §4, "$\lambda\not\subseteq\nu$"). If $\rho\subsetneq\kappa$, Lemma S of the Z file gives $\theta_{BR}:=D(\rho^\vee)\setminus D(\kappa^\vee)=R(\theta)$, $R(r,c)=(m-1-c,\,m-1-r)$. R is an involution of Δ_m, fixes the diagonal setwise, swaps rows and columns, and **reverses** the NW order. By Cor 5.4, $c_\rho=\mathcal{C}(\theta_{BR},1)$.

For a shifted skew shape φ let $K(\varphi) :=$ #labellings $\varphi \to \{1' < 1\}$ that strictly increase along every row (left to right) and every column (top to bottom), with no $1'$ on a diagonal box, and $K(\varphi) := 0$ if φ is not a rim. (BR: a rim has no two boxes with one strictly south and strictly east of the other.)

Lemma T (rim and chains)**.** *[proved] Let $\varphi = \nu/\lambda$ be a shifted skew shape. (a) If φ is not a rim, φ has a 3-chain in the NW order. (b) If φ is a rim, two boxes of φ are comparable iff they share a row or a column.*

Proof. (a) Take $a = (r, c)$, $b = (r', c') \in \varphi$ with $r < r'$, $c < c'$. Row ends of ν are weakly decreasing (ν strict, shifted), so $(r, c') \in \nu$. Rows of λ are initial segments of rows and $(r, c) \notin \lambda$, so $(r, c') \notin \lambda$. Then $a < (r, c') < b$. (b) Comparable means weakly NW; not strictly NW (rim) forces same row or column. □

Lemma 0.1. *[proved] For every $\rho \subsetneq \kappa$: $V(\theta) := V(\theta, \mathit{diagonal}) = K(\theta_{BR})$.*

Proof. If θ_{BR} is not a rim it has a 3-chain (Lemma T(a)); R reverses order, so θ has one, and $V(\theta) = 0 = K(\theta_{BR})$ (Lemma R). If θ_{BR} is a rim, by Lemma T(b) "strict along rows and columns" = "strict order-preserving" on θ_{BR}. Compose with R (order-reversing, diagonal-preserving) and rename $1 \mapsto e$, $1' \mapsto o$: strict order-preserving maps $\theta_{BR} \to \{1' < 1\}$ with diagonal $\mapsto 1$ correspond bijectively to strict order-preserving maps $\theta \to \{e < o\}$ with diagonal $\mapsto e$. □

So Q30 $\iff$ $\mathcal{C}(\varphi, 1) = T(\varphi) := (-1)^{|\varphi|-1} K(\varphi)$ for every nonempty shifted skew shape φ. (Then $c_\rho = (-1)^{|\theta|-1} V(\theta)$.)

3.2 BR Thm 5.5 at $p = 1$. BR Thm 5.5 (verbatim in `proofs/scratch-2026-10-24-q30-br-formula.md` §1b): $\mathcal{C}(\theta, p) = 0$ if θ is not a rim; $\mathcal{C}(\theta, p) = \delta_{\theta,\emptyset}$ for $p \le 0$; for a nonempty rim, $a = |\theta| - |\hat{\theta}|$ ($\hat{\theta} = \theta$ minus its north-east arm ψ):

(i) $\hat{\theta} = \emptyset$, θ meets the diagonal: column $\mapsto \delta_{p,|\theta|} - \delta_{p,|\theta|-1}$, row $\mapsto \delta_{p,|\theta|}$;
(ii) $\hat{\theta} = \emptyset$, θ off the diagonal: $2\delta_{p,|\theta|} - \delta_{p,|\theta|-1}$;
(iii) $\hat{\theta} \neq \emptyset$, ψ connected to $\hat{\theta}$: $\mathcal{C}(\hat{\theta}, p-a) - \mathcal{C}(\hat{\theta}, p-a+1)$;
(iv) $\hat{\theta} \neq \emptyset$, ψ not connected to $\hat{\theta}$: $a = 1$: $2\mathcal{C}(\hat{\theta}, p-1) - 2\mathcal{C}(\hat{\theta}, p)$; $a > 1$: $2\mathcal{C}(\hat{\theta}, p-a) - 3\mathcal{C}(\hat{\theta}, p-a+1) + \mathcal{C}(\hat{\theta}, p-a+2)$.

At $p = 1$, using $\mathcal{C}(\hat{\theta}, q) = 0$ for $q \le 0$ ($\hat{\theta} \neq \emptyset$):

(i) diagonal column: $|\theta| = 1 \mapsto 1$, $|\theta| = 2 \mapsto -1$, $|\theta| \ge 3 \mapsto 0$; diagonal row: $|\theta| = 1 \mapsto 1$, else 0.
(ii) off-diagonal row/column: $|\theta| = 1 \mapsto 2$, $|\theta| = 2 \mapsto -1$, $|\theta| \ge 3 \mapsto 0$.
(iii) $a = 1 \mapsto -\mathcal{C}(\hat{\theta}, 1)$; $a \ge 2 \mapsto 0$.
(iv) $a = 1 \mapsto -2\mathcal{C}(\hat{\theta}, 1)$; $a = 2 \mapsto +\mathcal{C}(\hat{\theta}, 1)$; $a \ge 3 \mapsto 0$.

3.3 Proof that $T(\varphi) = \mathcal{C}(\varphi, 1)$ (= BR Cor 5.6 at $p = 1$, which BR leave to the reader) [please referee]. Induction on $|\varphi|$. Non-rim: both sides 0. Let φ be a nonempty rim.

Geometry used throughout.

(g1) Rows and columns of a shifted skew shape are intervals.
(g2) A diagonal box (c, c) is the leftmost box of its row and the lowest box of its column (positions $(c, c' < c)$ and $(r > c, c)$ are not in the shifted plane).
(g3) Let r be the top row of φ and $B = (r, c)$ its rightmost box. Then c is the maximal column of φ: a box (r', c') with $c' > c$ has $r' > r$, and B would be strictly NW of it. So nothing of φ lies above or right of B, and the only possible neighbours of B are $L = (r, c-1)$ and $D = (r+1, c)$.
(g4) L and D are not both in φ (D is strictly SE of L). B is BR's "upper-right box"; the arm ψ is the largest row or column of the form $\varphi \cap$ (square with upper-right box B).
(g5) In a rim, comparable boxes share a row or column (Lemma T(b)), so $K(\varphi)$ only sees row and column constraints.

(i), (ii): $\varphi = \psi$ is a row or a column. Two labels, so $K = 0$ for length ≥ 3. Diagonal column (diagonal box at the bottom, g2): length 1: label 1, $K = 1$, $T = 1$; length 2: $1'$ over 1, $K = 1$, $T = -1$. Diagonal

row (diagonal box leftmost): length 1: $K = 1$, $T = 1$; length 2: the leftmost box would need $1'$, forbidden, $K = 0$. Off-diagonal: length 1: $K = 2$, $T = 2$; length 2: $K = 1$ $(1', 1)$, $T = -1$. All match §3.2 (i), (ii).

(iii) and (iv), $a \geq 3$**.** ψ is a row or column of ≥ 3 boxes, so $K(\varphi) = 0 = \mathcal{C}(\varphi, 1)$.

(iii), $a = 1$**.** $\psi = \{B\}$ and B is adjacent to $\hat{\theta}$, so exactly one of L, D is in φ (g3, g4). Let $S = (r+1, c-1)$. If $S \notin \varphi$, the 2×2 square at B meets φ in $\{L, B\}$ or $\{B, D\}$, a row or column of 2, so $a \geq 2$. Hence $S \in \varphi$, and $S \in \hat{\theta}$.

- $L \in \varphi$: L is directly above S, so every valid labelling of $\hat{\theta}$ has $L = 1'$ (column strict). B is right of L, so $B = 1$ is forced and allowed (unprimed). B's column contains only B ($D \notin \varphi$, g1). B's row: boxes left of L are also left of B, and they already satisfy their constraints with $L = 1'$ in $\hat{\theta}$ (a box left of L would need a label $< 1'$, impossible; then $K(\hat{\theta}) = 0$ and $K(\varphi) = 0$ too). So $K(\varphi) = K(\hat{\theta})$.
- $D \in \varphi$: S is directly left of D, so $D = 1$ in every labelling of $\hat{\theta}$. B above D forces $B = 1'$. B is off the diagonal: $D = (r+1, c)$ is in the shifted plane, so $c \geq r + 1$. B's row contains only B ($L \notin \varphi$). Again $K(\varphi) = K(\hat{\theta})$.

Then $T(\varphi) = (-1)^{|\hat{\theta}|} K(\hat{\theta}) = -T(\hat{\theta}) = -\mathcal{C}(\hat{\theta}, 1)$ by induction. ✓

(iii), $a = 2$**.** $\psi = \{L, B\}$ (a row) or $\{B, D\}$ (a column), and the 2×2 square meets φ in exactly ψ, so $S, D \notin \varphi$ (resp. $S, L \notin \varphi$). Row case: L's neighbours are $(r, c-2)$, S ($\notin$), and the box above L ($\notin$, r is the top row). B's other neighbour $D \notin \varphi$. Connection to $\hat{\theta}$ therefore goes through $(r, c-2) \in \varphi$: three boxes in row r, $K(\varphi) = 0$. Column case: D's neighbours are $(r+2, c)$, S ($\notin$) and $(r+1, c+1)$ ($\notin$, column c is maximal); B's other neighbour $L \notin \varphi$. So $(r+2, c) \in \varphi$: a column of three, $K(\varphi) = 0$. Both match $\mathcal{C} = 0$. ✓

(iv), $a \in \{1, 2\}$**:** ψ **not connected to** $\hat{\theta}$**.**

- No box of ψ shares a row or column with a box of $\hat{\theta}$: by (g1), walking along the common row/column interval from the ψ box to the $\hat{\theta}$ box we would cross a ψ–$\hat{\theta}$ adjacency.
- ψ contains no diagonal box. If ψ is a row in row r containing (r, r), then $\psi = \{(r, r), \ldots, (r, c)\}$ (g2), and any box $(r', c') \in \hat{\theta}$ has $r' > r$ (r is the top row), so $r < r' \leq c' \leq c$: column c' is a column of ψ. Contradiction. If ψ is a column in column c containing (c, c), then $\psi = \{(r, c), \ldots, (c, c)\}$ (g2), and any $(r', c') \in \hat{\theta}$ has $r \leq r' \leq c' \leq c$: row r' is a row of ψ. Contradiction. ($\hat{\theta} \neq \emptyset$.)
- By (g5) and the first bullet, the constraints on ψ and on $\hat{\theta}$ are independent: $K(\varphi) = K(\psi)K(\hat{\theta})$, with $K(\psi) = 2$ ($a = 1$, off-diagonal single box) or 1 ($a = 2$, off-diagonal domino: $1', 1$).
- $T(\varphi) = (-1)^{a+|\hat{\theta}|-1} K(\psi)K(\hat{\theta}) = (-1)^a K(\psi)T(\hat{\theta})$: $a = 1 \mapsto -2\mathcal{C}(\hat{\theta}, 1)$, $a = 2 \mapsto +\mathcal{C}(\hat{\theta}, 1)$. ✓

All cases of Thm 5.5 are covered ($\hat{\theta} = \emptyset$ iff φ is a row or column; otherwise (iii) or (iv)). □

Theorem Q30 ([proved given BR Cor 5.4, Thm 5.5, and Lemma S])**.** *$c_\rho = (-1)^{|\theta|-1} V(\theta)$ for all $\rho \subsetneq \kappa$, with V given by Lemma R: $c_\rho = 0$ iff θ has a 3-chain or a diagonal box (c, c) with $(c-1, c) \in \theta$; otherwise $|c_\rho| = 2^{\#\text{off-diagonal boxes of } \theta \text{ incomparable to every other box of } \theta}$.*

Remark. Route (i) of the scratch (Cor 5.6 at $p = 1$ read literally: content $\{1\}$ makes KLG (iii)/(iv) vacuous) gives the same statement in one line, but rests on a reader-left proof. §3.3 replaces it.

3.4 OG twin. I file Lemma 2 (OG): $c_\rho \neq 0 \Rightarrow \theta$ is a rook strip, and $c_\rho = (-1)^{|\theta|-1}$ for every rook strip $\theta = D(\kappa) \setminus D(\rho)$, from BR Lemma vanishB (a)–(c), proved in BR. For an up-set θ: rook strip $\iff$ antichain. ($\Rightarrow$: I file Lemma 2. $\Leftarrow$: two boxes in one row or column are comparable.) An up-set is an antichain iff it consists of maximal elements of P_κ, and every set of maximal elements is an up-set. So the OG operator is inclusion–exclusion over nonempty sets of corners, $|c_\rho| = 1$. □

4. Theorem $\mathrm{SR}_{\mathrm{LG}}$

For a finite poset P with $D \subseteq P$ and $n \geq 0$, let $N(P, D; n) = \#\{\text{strict order-preserving } f : P \to [n] \text{ with } f(D) \subseteq 2\mathbb{Z}\}$. Write $N_\kappa(j) := N(P_\kappa, \text{diagonal}; 2j-1)$.

Lemma 0.2 (peeling the top layer)**.** *[proved] For $j \geq 1$ and every κ: $N_\kappa(j+1) = N_\kappa(j) + \sum_{\rho \subsetneq \kappa} V(D(\kappa) \setminus D(\rho))\, N_\rho(j)$.*

Proof. Let $f : P_\kappa \to [2j+1]$ be strict order-preserving with even diagonal values. Put $\theta = f^{-1}\{2j, 2j+1\}$.

- θ is an up-set: $x \in \theta$, $x < y \Rightarrow f(y) > f(x) \geq 2j \Rightarrow f(y) = 2j+1$, so $y \in \theta$. Hence $\theta = D(\kappa) \setminus D(\rho)$ for a unique strict $\rho \subseteq \kappa$ (I file §2 Fact applied to the down-set $P_\kappa \setminus \theta$, an order ideal of Δ_m).
- $f|_\theta$ is strict order-preserving into $\{2j < 2j+1\}$ with diagonal boxes $\mapsto 2j$ (the only even value), i.e. a map counted by $V(\theta)$ under $e = 2j$, $o = 2j+1$.
- $f|_{D(\rho)}$ is strict order-preserving into $[2j-1]$ with even diagonal: counted by $N_\rho(j)$.

Conversely, given $\rho \subseteq \kappa$, a $V(\theta)$-map g and an $N_\rho(j)$-map h, glue $f = g \cup h$. Relations inside θ or inside $D(\rho)$ hold. For $x \in D(\rho)$, $y \in \theta$ with $x < y$: $f(x) \leq 2j-1 < 2j \leq f(y)$. The case $y < x$ with $y \in \theta$, $x \in D(\rho)$ is impossible (θ is an up-set). Diagonal values are even. The two constructions are inverse. $\rho = \kappa$ ($\theta = \emptyset$, $V = 1$) gives the term $N_\kappa(j)$. □

Theorem $\mathbf{SR}_{\mathrm{LG}}$ ([proved given (G0), (G1), (BR)]). *For every strict κ with parts in $[m]$ and every integer $j \geq 1$,*

$$(-1)^{|\kappa|} H_\kappa(-j) = N_\kappa(j) = \#\{\textit{strict order-preserving } P_\kappa \to [2j-1] \textit{ with even values on diagonal boxes}\}.$$

Proof. Induction on j, simultaneously for all κ. $j = 1$: $[1]$ has no even value, and every nonempty P_κ contains the diagonal box $(0,0)$, so $N_\kappa(1) = \delta_{\kappa,\emptyset} = s_1(\kappa)$ (anchor). Step: by Theorem Q30, $|c_\rho| = V(\theta)$ for $\rho \subsetneq \kappa$ and $c_\rho = 0$ otherwise, so $(\star)$ and Lemma 0.2 are the same recursion: $s_{j+1}(\kappa) = s_j(\kappa) + \sum_{\rho \subsetneq \kappa} V(\theta) s_j(\rho)$, and the same for N. □

Hand check. $\kappa = (2,1)$: $P = \{(0,0) < (0,1) < (1,1)\}$, a 3-chain with both ends diagonal. $N(j) = \#\{2 \leq x < y < z \leq 2j-1,\ x, z \text{ even}\}$: $j = 3$ ([5]): $(2,3,4)$ only, so 1. $(\star)$: the Pieri support of $(2,1)$ is $\{\rho = (2),\ \theta = \{(1,1)\},\ V = 1\}$ (the others have $V = 0$: $\rho = (1)$ has $\theta = \{(0,1),(1,1)\}$, diagonal non-minimal; $\rho = \emptyset$ is a 3-chain). $s_3(2,1) = s_2(2,1) + s_2(2) = 0 + 1 = 1$ ✓ ($s_2(2) = N_{(2)}(2) = \#\{2 \leq x < y \leq 3,\ x \text{ even}\} = 1$). $\kappa = (1)$: $N(j) = \#\text{even in } [2j-1] = j-1$, $H_{(1)}(x) = x+1$ ✓. $\kappa = (2)$: $N(j) = \sum_{x \text{ even} \leq 2j-2} (2j-1-x) = (j-1)^2$, $H_{(2)}(x) = (x+1)^2$ ✓.

5. Theorem $\mathrm{SR}_{\mathrm{OG}}$

Lemma 0.3. *[proved] For $n \geq 0$: $\bar{\Omega}(P_\kappa, n+1) = \bar{\Omega}(P_\kappa, n) + \sum_{\emptyset \neq A \subseteq \max(P_\kappa)} \bar{\Omega}(P_\kappa \setminus A, n)$, and $P_\kappa \setminus A = D(\rho_A)$.*

Proof. The preimage of the top value $n+1$ under a strict order-preserving map is an up-set (as in 0.2) and an antichain (strictness), i.e. a set A of maximal elements; the rest maps strictly into $[n]$. Gluing is as in 0.2. □

Theorem $\mathbf{SR}_{\mathrm{OG}}$ ([proved given (G0), (G1), (BR)]). $(-1)^{|\kappa|} H^{\mathrm{OG}}_\kappa(-j) = \bar{\Omega}(P_\kappa, j-1)$ *for all $j \geq 1$.*

Proof. $j = 1$: $\bar{\Omega}(P, 0) = \delta_{P,\emptyset} = s_1$ (anchor, I file Lemma 1, both types). Step: $(\star)$ on OG with $|c_\rho| = 1$ exactly on $\rho = \rho_A$, $A \neq \emptyset$ a set of corners (§3.4), is Lemma 0.3 with $n = j-1$. □

(This is Stanley reciprocity for the order polynomial $H^{\mathrm{OG}}_\kappa(j) = \Omega(P_\kappa, j+1)$, the SMT model of the I file §9, but proved without SMT.)

6. Corollary: Theorem I, combinatorially

Let $r = r(\kappa)$ be the longest chain of P_κ; by I file Lemma 0 it is realised by a staircase path starting at $(0,0)$ and ending at a box with $a+b = r-1$, and $\max(a+b) = r-1$.

- **LG.** $N_\kappa(j) > 0 \iff 2j-1 \geq r+1 \iff j \geq \lceil r/2 \rceil + 1$. ($\Leftarrow$) $f(a,b) = a+b+2$ is strict order-preserving ($a+b$ strictly increases along $x < y$), takes values in $[2, r+1]$, and $f(a,a) = 2a+2$ is even. ($\Rightarrow$) Along the staircase chain, $f((0,0)) \geq 2$ (even) and f increases strictly over r boxes, so the last value is $\geq r+1$. Since $2j-1$ is odd: r even $\Rightarrow j \geq r/2+1$; r odd $\Rightarrow 2j-1 \geq r+2 \Rightarrow j \geq (r+3)/2 = \lceil r/2 \rceil + 1$.
- **OG.** $\bar{\Omega}(P_\kappa, j-1) > 0 \iff j-1 \geq r$ (a chain of r needs r values; $f(a,b) = a+b+1$ works).
- Monotonicity in j is inclusion of alphabets ($[2j-1] \subset [2j+1]$ preserving parity).

With SR this re-proves Theorem I (zero set exactly $\{-1,\dots,-W\}$, sign $(-1)^{|\kappa|}$ beyond, s_j nondecreasing). The inputs are the same (G0), (G1), (BR) plus the full Q30, so this is a second proof, not a weaker one.

7. Verification

- `prove254/srcheck.py m`, $m = 2..5$: $H_\kappa(-j)$ computed **directly from Demazure characters at negative weights** (`prove249d/dem.py` type C; `wake253/og.py` type B), $j = 1..4$, against brute-force strict counts. LG and OG: 16/16, 32/32, 64/64, 128/128 (κ, j) pairs, 0 failures. Independent of $(\star)$, of BR, and of the LS/SMT models.
- Negative controls at $m = 4$ (64 pairs): LG alphabet $[2j]$ instead of $[2j-1]$: 34 failures; LG without the diagonal parity: 34 failures; OG alphabet $[j]$: 17 failures. The test separates.
- Wake 254: Q30 1772/1772 nonzero coefficients $m \le 7$, support exact on 8532 pairs (`q30.py`); `brcheck.py`: BR eq. (11) = literal KLG count = Lemma R closed form on all 2225 shifted skew shapes $2 \le n \le 6$.

Scripts: `work-in-progress/scripts/prove254/` and `work-in-progress/scripts/wake254/q30/` (WIP f17e0cf).

8. Gaps

(1) (G0), (G1): standard, cited, not deep-read (same debt as Theorems Z, I, C).
(2) (BR): Cor 5.4, Thm 5.5 (LG) and Lemma vanishB (OG) read first-hand, proved in BR. Cor 5.6 is NOT used.
(3) §3.3 is a case analysis I wrote and cold re-read once. It has not been refereed by anyone else.
(4) Nothing else.

9. The half-integers: the whole odd-zero law, with signs, from one row

PROVE.md's stretch (Q31): iterate $(\star)$ from the half-integer anchor $H(-\frac{1}{2}) = 2^{-\kappa_1}$ and look for positivity. It went through. Notation: $s_x(\kappa) := (-1)^{|\kappa|} H_\kappa(-x)$ for $x \in \frac{1}{2}\mathbb{Z}$; $\kappa' :=$ κ minus its largest part; $r' := r(\kappa')$; $M := 2a-1$; $n := \kappa_1$.

Theorem HZ ([proved given (G0), (G1), (BR)]). *For every $\kappa \neq \emptyset$ and every integer $a \ge 1$:*

- *(small a) if $2a \le \kappa_1$: $s_{a+\frac{1}{2}}(\kappa) = (-1)^{\kappa_1} 2^{-\kappa_1} N_{\kappa'}(a)$;*
- *(large a) if $2a \ge \kappa_1$: $s_{a+\frac{1}{2}}(\kappa) = \sum_g \Phi(u(g))$ with every $\Phi(u(g)) \ge 2^{-\kappa_1}$, so $s_{a+\frac{1}{2}}(\kappa) \ge 2^{-\kappa_1} N_{\kappa'}(a)$;*
- *hence $H_\kappa(-a-\frac{1}{2}) = 0 \iff N_{\kappa'}(a) = 0 \iff \kappa' \neq \emptyset$ and $a \le \lceil r'/2 \rceil$ (§6), and otherwise $\operatorname{sgn} H_\kappa(-a-\frac{1}{2}) = (-1)^{|\kappa|-\kappa_1}$ if $2a \le \kappa_1$, $(-1)^{|\kappa|}$ if $2a \ge \kappa_1$ (the two agree at $2a = \kappa_1$, κ_1 even).*

In the odd-zero-law notation ($H(-M/2)$, M odd): zeros exactly at $M = 3, 5, \dots, 2\lceil r'/2 \rceil + 1$, none for single rows. That is the **odd-zero law, both halves**: (Z) (Theorem Z had it by another induction) and (NZ) (open since Wake 251; PROVE 253 reduced it to an odd-μ_1 base inequality, which is now moot), **with strictness and signs**. The small-a bullet is Prop. R of the odd-zero-law file, re-proved here without the residue calculus.

Inputs: $(\star)$ as a polynomial identity (it holds at all integers, both sides are polynomials, so it holds at every x), Theorem Q30, Thm A of `2026-10-23-odd-zero-law.md` ($H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, proved given (G1)), and §6.

9.1 Layer operators. For up-sets $\theta = D(\kappa) \setminus D(\nu)$ put $A_o(\kappa,\nu) = [\theta$ is an antichain with no diagonal box$]$, $A_e(\kappa,\nu) = [\theta$ is an antichain$]$.

Lemma 0.4. *[proved] (a) $A_o A_e = I + C$, where $C(\kappa,\nu) = |c_\nu|$. (b) For $a \ge 1$, $[(A_o A_e)^{a-1} A_o](\kappa,\nu) = N_{\kappa/\nu}(a) := \#\{$strict order-preserving $D(\kappa) \setminus D(\nu) \to [2a-1]$, even values on diagonal boxes$\}$.*

Proof. (b) Peel values from the top, exactly as in Lemma 0.2: the preimage of the current top value is an antichain up-set of what remains; the value is odd (no diagonal box allowed: A_o) or even (A_e). The values $2a-1, 2a-2, \dots, 1$ alternate $o, e, \dots, o$, giving $A_o(A_e A_o)^{a-1} = (A_o A_e)^{a-1} A_o$. Gluing is as in 0.2. (a) The same with the two-letter alphabet $\{e < o\}$: $(A_o A_e)(\kappa,\nu) = V(\theta)$ ($\theta = \emptyset$ gives 1), and $V(\theta) = |c_\nu|$ for $\nu \subsetneq \kappa$ by Q30. □

So $(\star)$ reads $s_{x+1} = A_o A_e\, s_x$ for every $x \in \mathbb{C}$.

Lemma 0.5 (the anchor collapses to single rows)**.** *[proved]* $\varepsilon := A_e\, s_{\frac12}$ *satisfies* $\varepsilon(\emptyset) = 1$, $\varepsilon((L)) = (-1)^{L+1}2^{-L}$ *for* $L \geq 1$, *and* $\varepsilon(\kappa) = 0$ *if* $\ell(\kappa) \geq 2$.

Proof. $s_{\frac12}(\nu) = (-1)^{|\nu|}2^{-\nu_1}$ (Thm A). $\varepsilon(\kappa) = \sum_{A \subseteq \max P_\kappa} s_{\frac12}(\kappa \setminus A)$. If $\ell(\kappa) \geq 2$, the last box z of the bottom row is maximal and not in row 0; toggling $z \in A$ changes $|\nu|$ by one and leaves ν_1 alone, so the terms cancel in pairs. If $\kappa = (L)$, the only maximal box is $(0, L-1)$: $\varepsilon = (-1)^L 2^{-L} + (-1)^{L-1}2^{-(L-1)} = (-1)^{L+1}2^{-L}$. □

Corollary 0.6 (row formula)**.** *[proved] For* $a \geq 1$*:*

$$s_{a+\frac12}(\kappa) = N_\kappa(a) + \sum_{L=1}^{\kappa_1} (-1)^{L+1}2^{-L} N_{\kappa/(L)}(a),$$

where $\kappa/(L) = P_\kappa$ *minus the first* L *boxes of row 0.*

Proof. $s_{a+\frac12} = (A_oA_e)^a s_{\frac12} = (A_oA_e)^{a-1}A_o\,\varepsilon$, then Lemmas 0.4(b) and 0.5. □

(Example $\kappa = (3)$, $a = 2$: $0 + \frac12 \cdot 3 - \frac14 \cdot 3 + \frac18 \cdot 1 = \frac78 = s_{5/2}(3)$.)

9.2 Fibre over the rows below row 0. Rows ≥ 1 of P_κ form a copy of $P_{\kappa'}$ ($(i,j) \mapsto (i-1, j-1)$ preserves order and the diagonal). Row 0 is the chain $(0,0) < \cdots < (0, n-1)$; its only relations to rows ≥ 1 are $(0,\beta) < (1,\beta)$ for $1 \leq \beta \leq \kappa_2$, and nothing in rows ≥ 1 lies below a row-0 box. Fix g : rows $\geq 1 \to [M]$ strict order-preserving with even diagonal (there are $N_{\kappa'}(a)$ of them) and put $u_\beta := g(1,\beta)$ ($1 \leq \beta \leq k := \kappa_2 \leq n-1$); $u_\beta := M+1$ (no bound) for $k < \beta \leq n-1$. For a fixed bound vector u let

- $G_L(M;u) := \#\{\text{strictly increasing } v_L < \cdots < v_{n-1} \text{ in } [M] \text{ with } v_\beta < u_\beta\}$, for $1 \leq L \leq n$ ($G_n := 1$);
- $E_0(M;u) :=$ the same for $L = 0$ with v_0 even;
- $\Phi(u) := E_0 - \sum_{L=1}^n (-\frac12)^L G_L$.

Corollary 0.6 sums over g: $\boxed{s_{a+\frac12}(\kappa) = \sum_g \Phi(u(g))}$. Bounds from a real g satisfy **(H)**: $u_\beta \geq \beta + 1$ for the bounded boxes $1 \leq \beta \leq k$ (unbounded boxes are handled in the proof of 0.8), because $u_1 = g(1,1)$ is even (≥ 2) and row 1 is strictly increasing.

9.3 The one-row lemma [please referee]. Let $\Psi_L(M;u) := \sum_{L'=L}^n (-\frac12)^{L'-L} G_{L'}(M;u)$, so $\Psi_n = 1$ and $\Phi = E_0 + \frac12 \Psi_1$.

Lemma 0.7 (shift)**.** *[proved] Let* $1 \leq L \leq n-1$, $M \geq 1$, *and* $u_{L'} \geq 2$ *for* $L \leq L' \leq n-1$. *Then*

$$\Psi_L(M;u) = G_L(M-1; u-1) + \tfrac12 \Psi_{L+1}(M-1; u-1).$$

Proof. For $L' \geq L$ let $a_{L'}$ (resp. $b_{L'}$) count the fillings of boxes $L'..n-1$ with $v_{L'} = 1$ (resp. ≥ 2); $b_n := 1$. Then $G_{L'} = a_{L'} + b_{L'}$ ($L' \leq n-1$). Deleting box L' (value 1) is a bijection onto the fillings of $L'+1..n-1$ with all values ≥ 2; its inverse puts 1 at box L', which is allowed because $u_{L'} \geq 2$ and $M \geq 1$. So $a_{L'} = b_{L'+1}$ for $L \leq L' \leq n-1$ (for $L' = n-1$ both are 1). Substituting,

$$\begin{aligned}\Psi_L &= \sum_{L'=L}^{n-1} (-\tfrac12)^{L'-L}(b_{L'} + b_{L'+1}) + (-\tfrac12)^{n-L} b_n = b_L + \sum_{L''=L+1}^{n} b_{L''}(-\tfrac12)^{L''-L-1}(1-\tfrac12) \\ &= b_L + \tfrac12 \sum_{L''>L} (-\tfrac12)^{L''-L-1} b_{L''}.\end{aligned}$$

Finally $v \mapsto v-1$ identifies the fillings with values in $[2, M]$ and bounds u with fillings in $[M-1]$ and bounds $u-1$: $b_{L''}(M;u) = G_{L''}(M-1; u-1)$ (also for $L'' = n$). □

Lemma 0.8 (one row)**.** *[proved] Assume (H). (a) If $M \geq n-1$:*

$$\Phi(u) = E_0(M;u) + \sum_{i=0}^{n-2} 2^{-i-1} G_{1+i}(M-1-i;\, u-1-i) + 2^{-n} \;\geq\; 2^{-n}.$$

(b) If $M \leq n-1$: $\Phi(u) = (-1)^{n-1-M} 2^{-n}$.

Proof. Apply Lemma 0.7 repeatedly: step i ($i = 0, 1, \dots$) rewrites $\Psi_{1+i}(M-i; u-i)$. Its hypotheses are $M - i \geq 1$ and $u_{L'} - i \geq 2$ for $L' \geq 1+i$; the second holds by (H) ($u_{L'} - i \geq L' + 1 - i \geq 2$), also for unbounded boxes while $M - i \geq 1$. (a) $M \geq n-1$ allows $i = 0..n-2$, ending at $\Psi_n = 1$: $\Psi_1 = \sum_{i=0}^{n-2} 2^{-i} G_{1+i}(M-1-i; u-1-i) + 2^{-(n-1)}$. Every term is ≥ 0. (b) $M \leq n-1$ allows $i = 0..M-1$, ending at $\Psi_{1+M}(0; u-M) = \sum_{L'} (-\frac{1}{2})^{L'-1-M} G_{L'}(0) = (-\frac{1}{2})^{n-1-M}$ (only $G_n = 1$ survives an empty alphabet). The G-terms vanish: $G_{1+i}(M-1-i)$ fills $n-1-i$ boxes from $M-1-i < n-1-i$ values. So $\Psi_1 = 2^{-M}(-\frac{1}{2})^{n-1-M}$. $E_0 = 0$ (n values ≥ 2 need $M \geq n+1$). Then $\Phi = \frac{1}{2}\Psi_1$. □

9.4 Proof of Theorem HZ. $s_{a+\frac{1}{2}}(\kappa) = \sum_g \Phi(u(g))$ (§9.2), the bounds satisfy (H), $M = 2a-1$. If $2a \leq \kappa_1$, then $M \leq n-1$ and Lemma 0.8(b) gives $\Phi \equiv (-1)^{n-2a} 2^{-n} = (-1)^{\kappa_1} 2^{-\kappa_1}$. If $2a \geq \kappa_1$, then $M \geq n-1$ and Lemma 0.8(a) gives $\Phi \geq 2^{-n}$. The number of g is $N_{\kappa'}(a)$ (with $N_\emptyset = 1$), and §6 applied to κ' says $N_{\kappa'}(a) = 0$ iff $\kappa' \neq \emptyset$ and $a \leq \lceil r'/2 \rceil$. □

Worked example ($\kappa = (3,2)$, $a = 2$, $M = 3 \geq n-1 = 2$). Rows ≥ 1: $(1,1) < (1,2)$, $(1,1)$ even, values in $[3]$: only $g = (2,3)$, so $u = (2,3)$. $E_0 = 0$ ($v_1 < 2$ is impossible above an even v_0). $i = 0$: $G_1(2;(1,2)) = 0$ ($v_1 < 1$). Terms: $0 + 0 + 0 + \frac{1}{8}$. So $s_{5/2}(3,2) = \frac{1}{8}$, which is the value ($\star$) gives, and it equals $2^{-\kappa_1} N_{(2)}(2)$: the bound is attained.

9.5 Verification.

- `prove254/halfthm.py m 5`: $(-1)^{|\kappa|} H_\kappa(-a-\frac{1}{2})$ computed from **Demazure characters directly** (type C, interpolated to $x = -a - \frac{1}{2}$), against $\sum_g \Phi(u(g))$, all κ with parts $\leq m$, $a = 1..5$. Also checks Lemma 0.8 pointwise for every g. $m = 2$: 15/15; $m = 3$: 35/35; $m = 4$: 75/75; Lemma 0.8 pointwise 477/477, 0 failures. ($m = 5$ Demazure run: not run to completion, not counted.)
- `prove254/hz_vs_rec.py 6 4`: Theorem HZ's closed forms against the ($\star$)-recursion values, all κ with parts ≤ 6, $a = 1..4$: 252/252. (Checks the algebra of §9, not the inputs.)
- `prove254/rowform.py 5 5`: row formula (Cor. 0.6) = ($\star$)-recursion values, 155/155.
- `prove254/rowphi*.py`: Lemma 0.8 needs (H). Without it Φ goes negative ($u = (1)$: $-\frac{1}{4}$; $u = (2,2)$, weakly increasing: $-\frac{1}{8}$), so the hypothesis is load-bearing and the test separates. With $M \leq n-2$, $\Phi < 0$ is the predicted sign $(-1)^{n-1-M}$.
- Data (`half.py`): $2^{\kappa_1} s_{a+\frac{1}{2}}(\kappa)$ is always an integer, e.g. $(5,4)$ at $a = 3$ gives 1, the minimum $2^{-\kappa_1} N_{\kappa'}$.

9.6 What this closes.

- (NZ) at half-integers: all κ, all a, with strictness. PROVE 253's open base inequality (odd μ_1) is not needed.
- Prop. R (odd-zero-law file) gets a second, residue-free proof (§9.4 small-a case).
- (Z) gets a second proof (zero iff $N_{\kappa'}(a) = 0$), independent of Theorem Z's induction.
- Consequence: Thm B ((OC) for Schubert varieties) no longer needs Brion–Kumar G2–G4.
- By-product [computed]: $T := (I+C)^{1/2}$ (so that $s_{x+\frac{1}{2}} = T s_x$) has entries $0, \pm 2^{-k}$. It is multiplicative over connected components of θ; an off-diagonal singleton has weight 1, an off-diagonal component of size ≥ 2 has weight 0, and a component containing exactly one diagonal box is a ribbon D with weight $\pm 2^{-|D|}$ (sign + iff its NE end box has a box to its left). Computed $m \leq 6$ (1716/1716 pairs, `tprime.py`). Not proved, not needed.

10. Gaps (updated)

Theorem HZ adds only Thm A (proved given (G1), odd-zero-law file) to the inputs of §8. §9 is self-contained otherwise. Lemmas 0.7/0.8 were written once and re-read once; nobody else has refereed them.

11. Addendum (Dream 254): the models are corollaries, and a question for you

- **Models come for free** [sketched]. $\mathrm{SR}_{\mathrm{LG}}$ + Ehrhart–Macdonald reciprocity on the order polytope $\mathcal{O}_P$ gives $H_\kappa(j) = F_\kappa(2j)$ for all j, with the lattice $\mathbb{Z}$ on diagonal coordinates and $\frac{1}{2}\mathbb{Z}$ off the diagonal, and F_κ the even-diagonal order-polynomial count. So the Wake 251 LS-parity model is a corollary (not yet written into the proof file). The OG Ω-model follows the same way from $\mathrm{SR}_{\mathrm{OG}}$ + Stanley reciprocity.
- **Q32 (question for you).** Relabel odd values as primed letters. Then "diagonal even" becomes "no primes on the diagonal", and $\mathrm{SR}_{\mathrm{LG}}$ reads as a strict marked-tableau count. Have you seen $H_\kappa(-j)$ for LG Schubert varieties stated as a count of strict fillings with even diagonal, i.e. an enriched P-partition count (Stembridge), or in De Concini's symplectic standard tableaux? If it is in print, I cite it and the write-up shrinks.